Object Oriented Analysis and Design Using the UML Version 4.2

Module 7C: Use-Case Analysis

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Objectives: Use-Case Analysis

- Understand the purpose of Use-Case Analysis and where in the lifecycle it is performed
- Identify the classes which perform a use-case flow of events
- Distribute the use-case behavior to those classes, identifying responsibilities of the classes
- Develop use-case realizations that model the collaborations between instances of the identified classes

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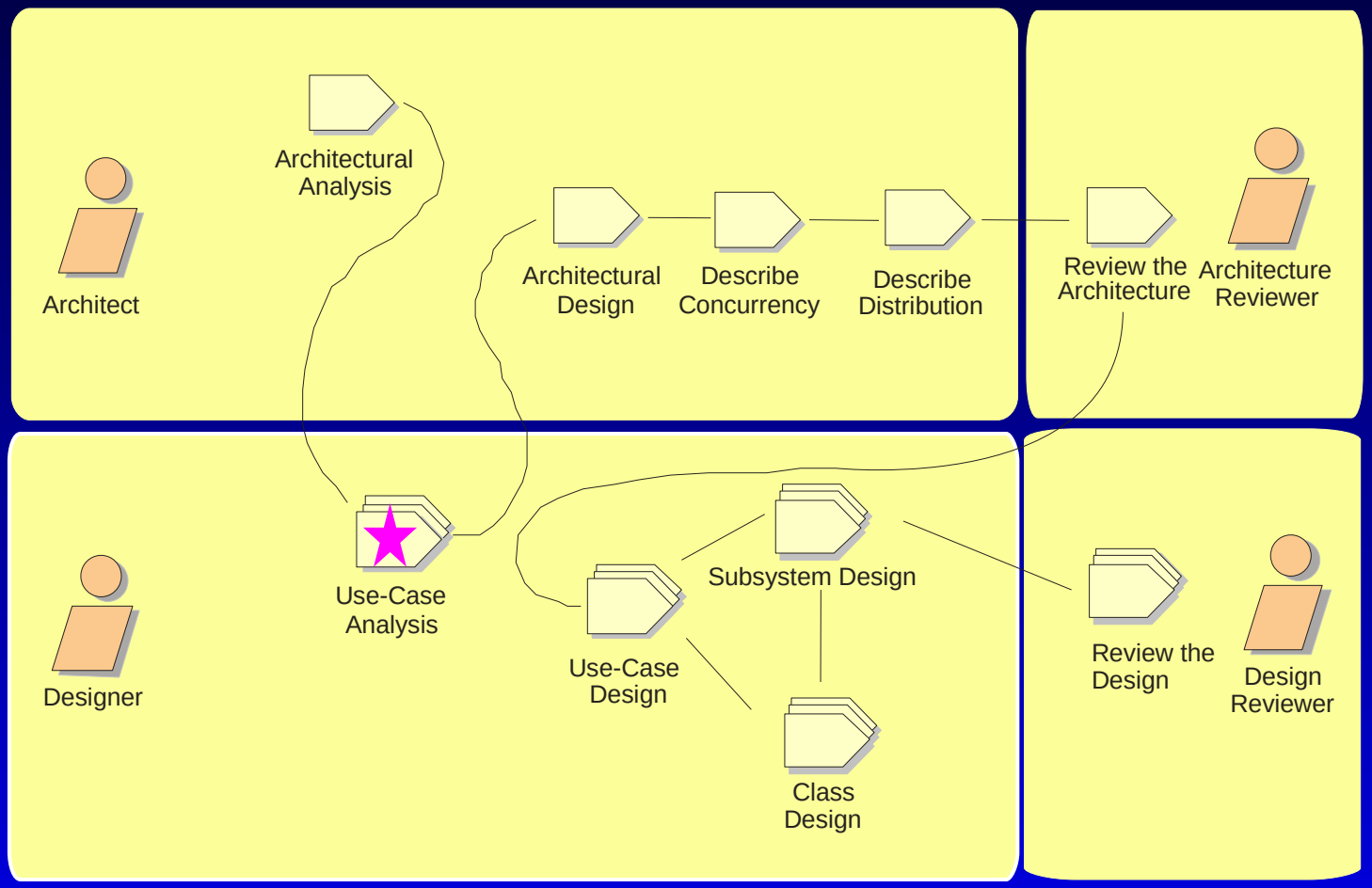
Use-Case Analysis in Context



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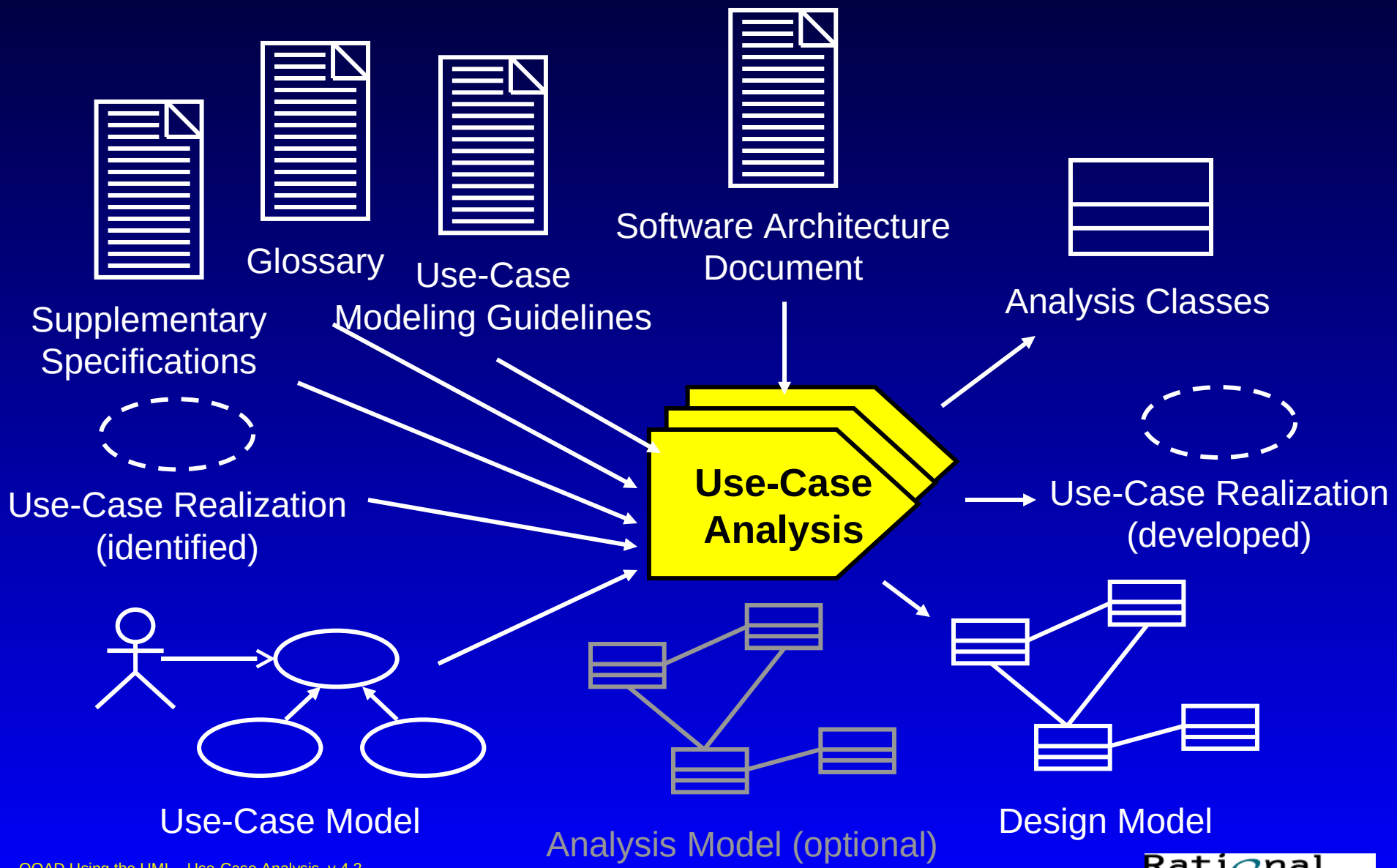
Use-Case Analysis Overview



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Use-Case Analysis Steps

- Supplement the Use-Case Description
- For each use-case realization
  - Find Classes from Use-Case Behavior
  - Distribute Use-Case Behavior to Classes
- For each resulting analysis class
  - Describe Responsibilities
  - Describe Attributes and Associations
  - Qualify Analysis Mechanisms
- Unify Analysis Classes
- Checkpoints

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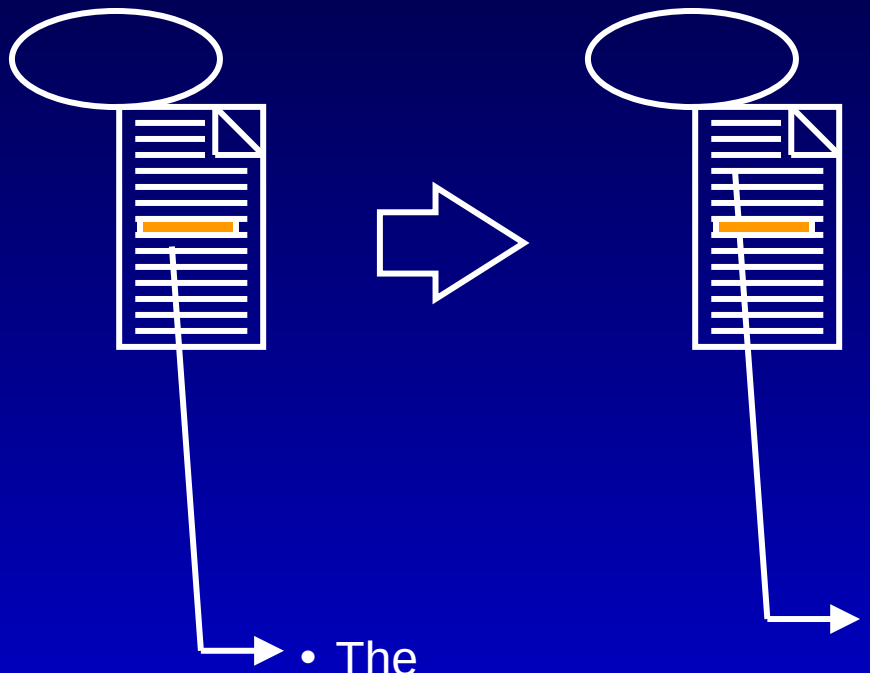
Supplement the Use-Case Description



- The system displays a list of course offerings.

- The system retrieves and displays a list of current course offerings from the course catalog legacy database.

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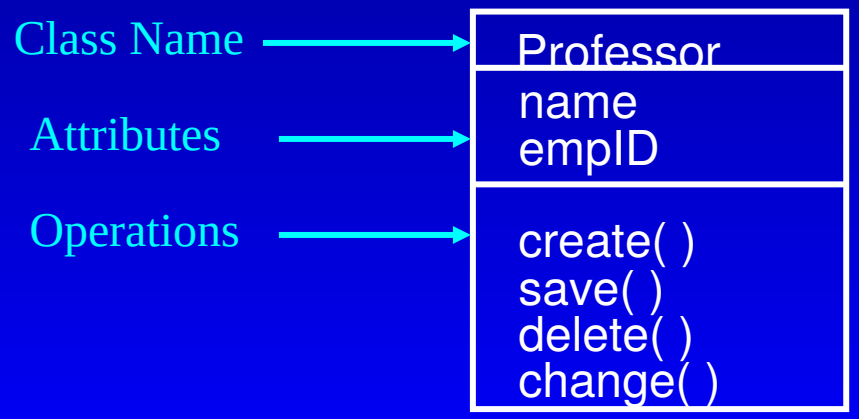
Review: Class

- An abstraction
- Describes a group of objects with common:
  - Properties (attributes)
  - Behavior (operations)
  - Relationships
  - Semantics



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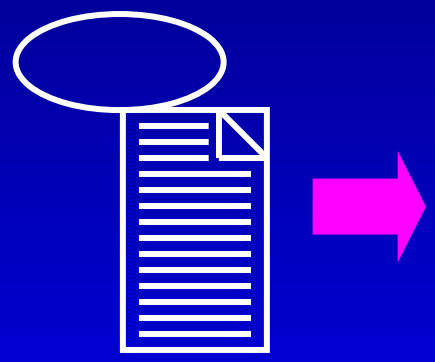
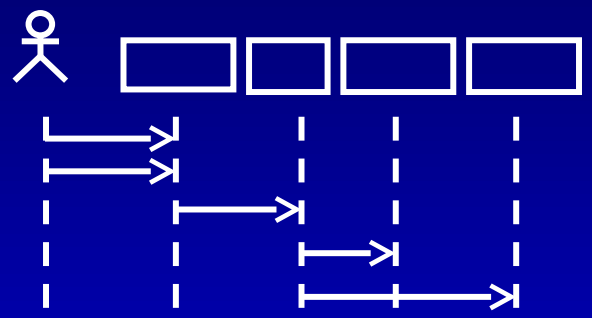
Review: Use-Case Realization

Use-Case Model *Design Model*

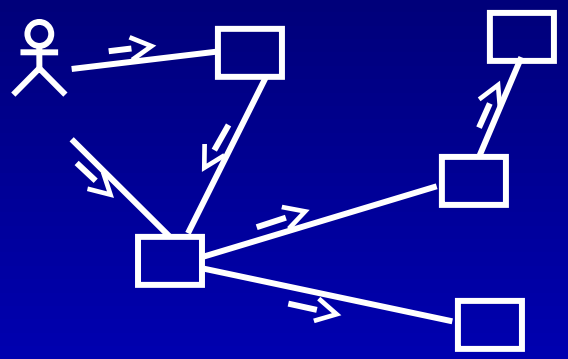
Use Case Use-Case Realization



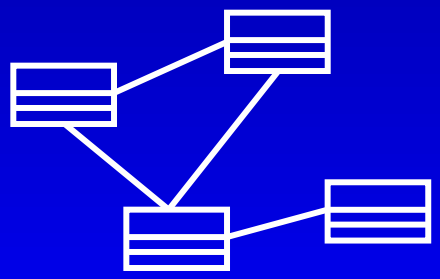
Use Case



Sequence Diagrams



Collaboration Diagrams



Class Diagrams

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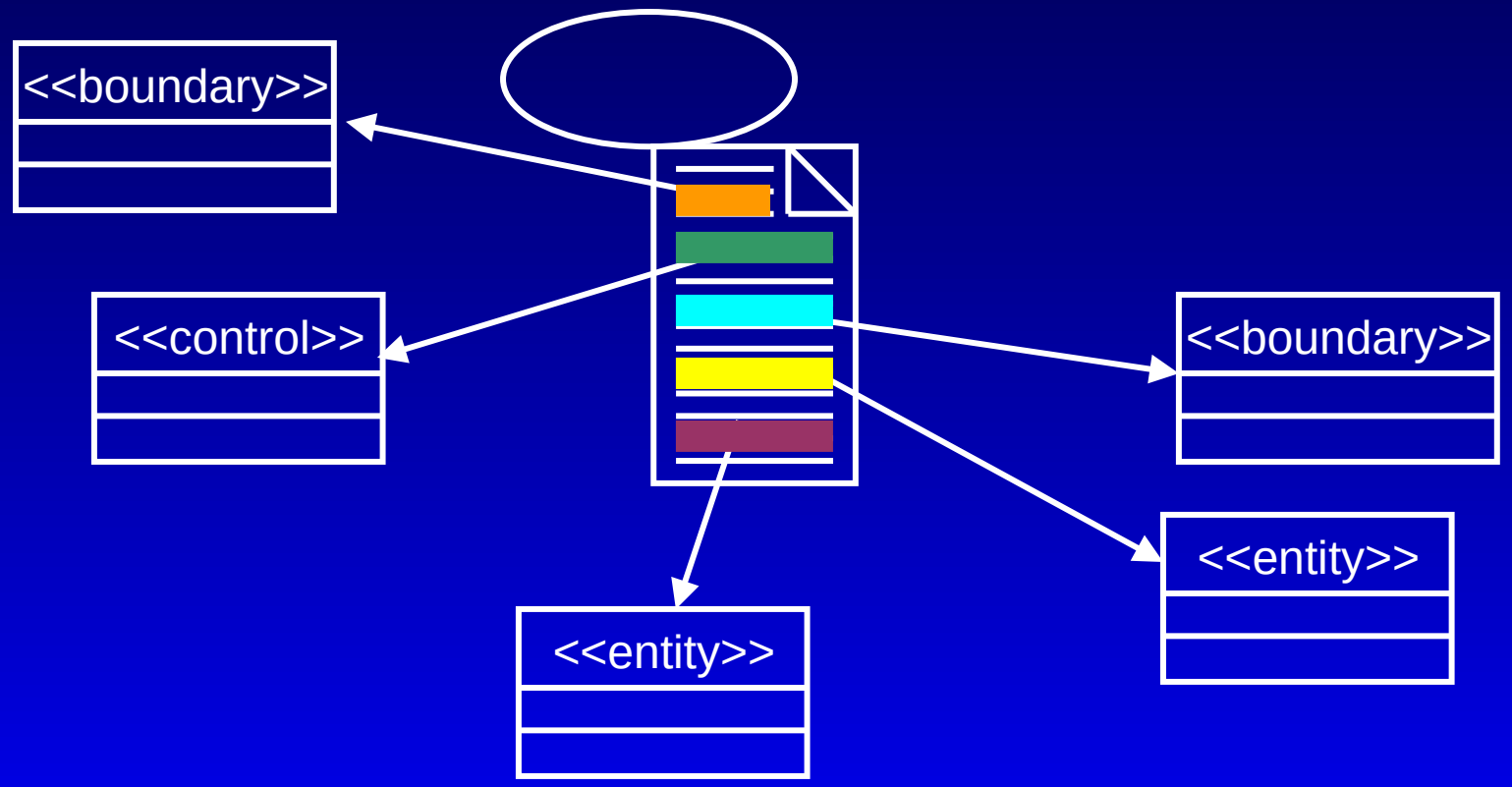
Find Classes From Use-Case Behavior

- The complete behavior of a use case has to be distributed to analysis classes



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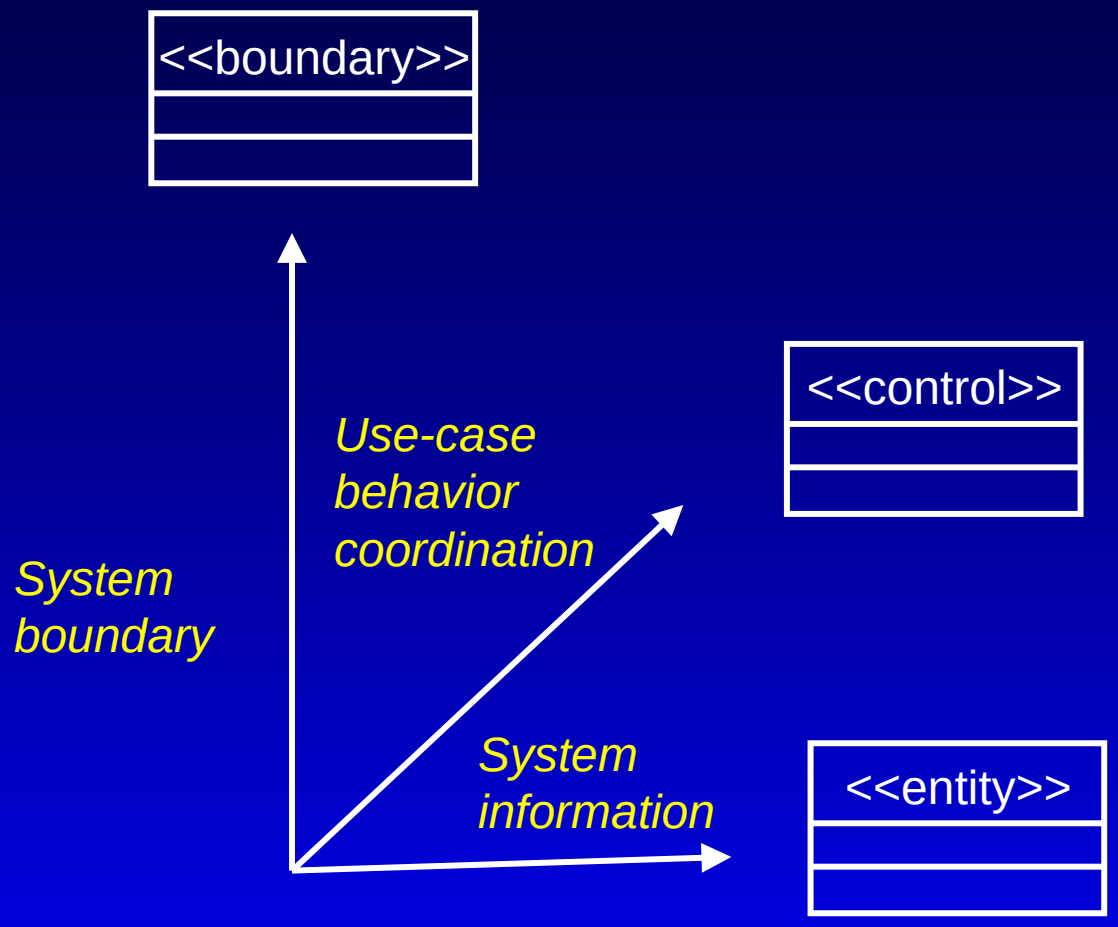
What is an Analysis Class?



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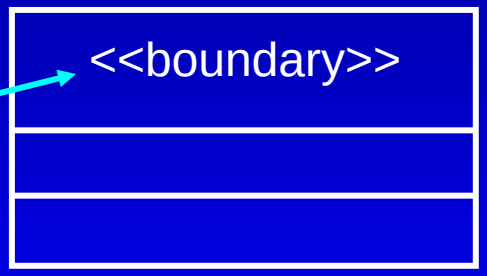
What is a Boundary Class?

- Intermediates the interface to something outside the system
- Several Types
  - User interface classes
  - System interface classes
  - Device interface classes
- *One boundary class per actor/use case pair*

Analysis class stereotype



Environment Dependent

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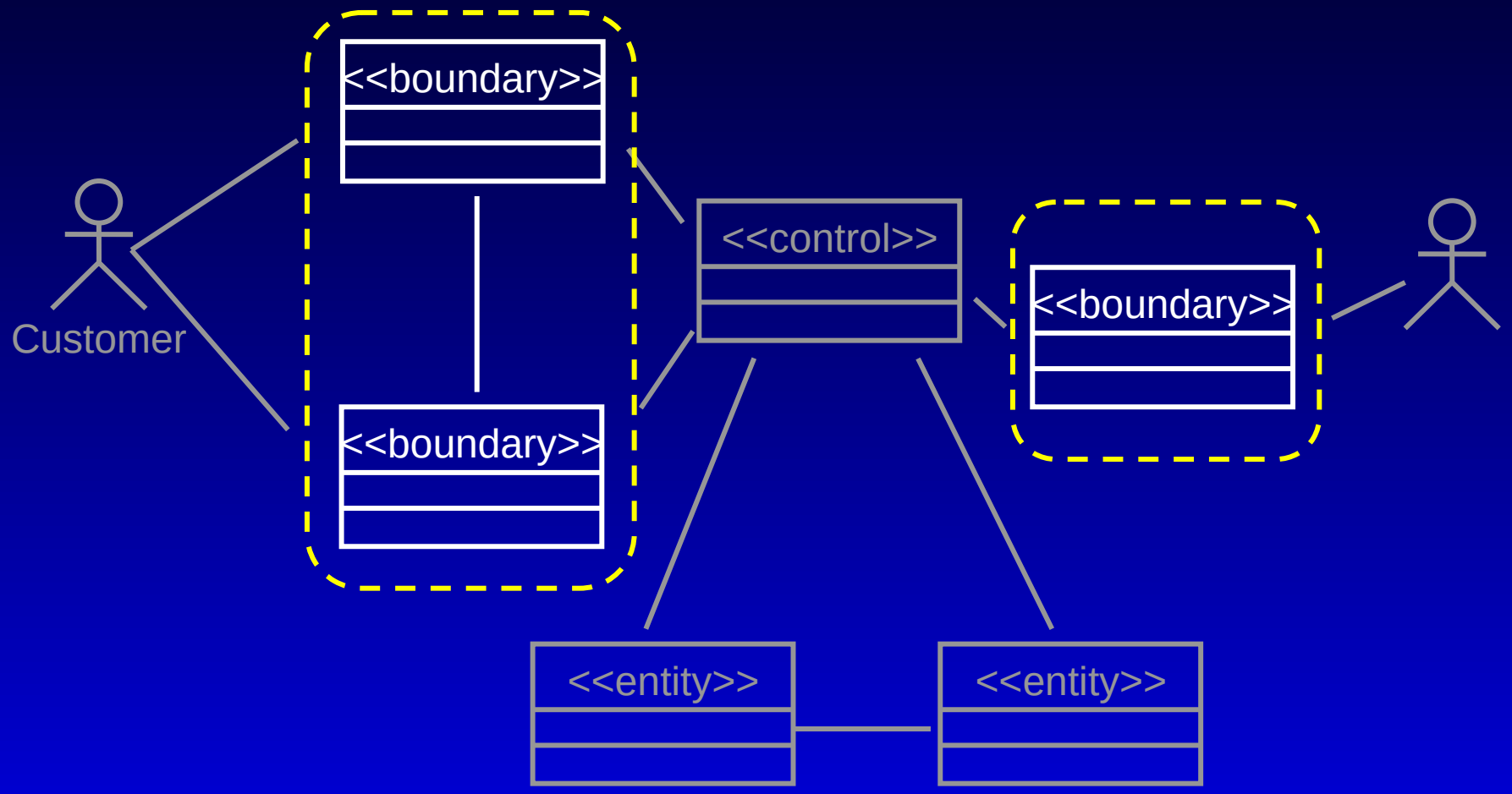
The Role of a Boundary Class



Model interaction between the system and its environment

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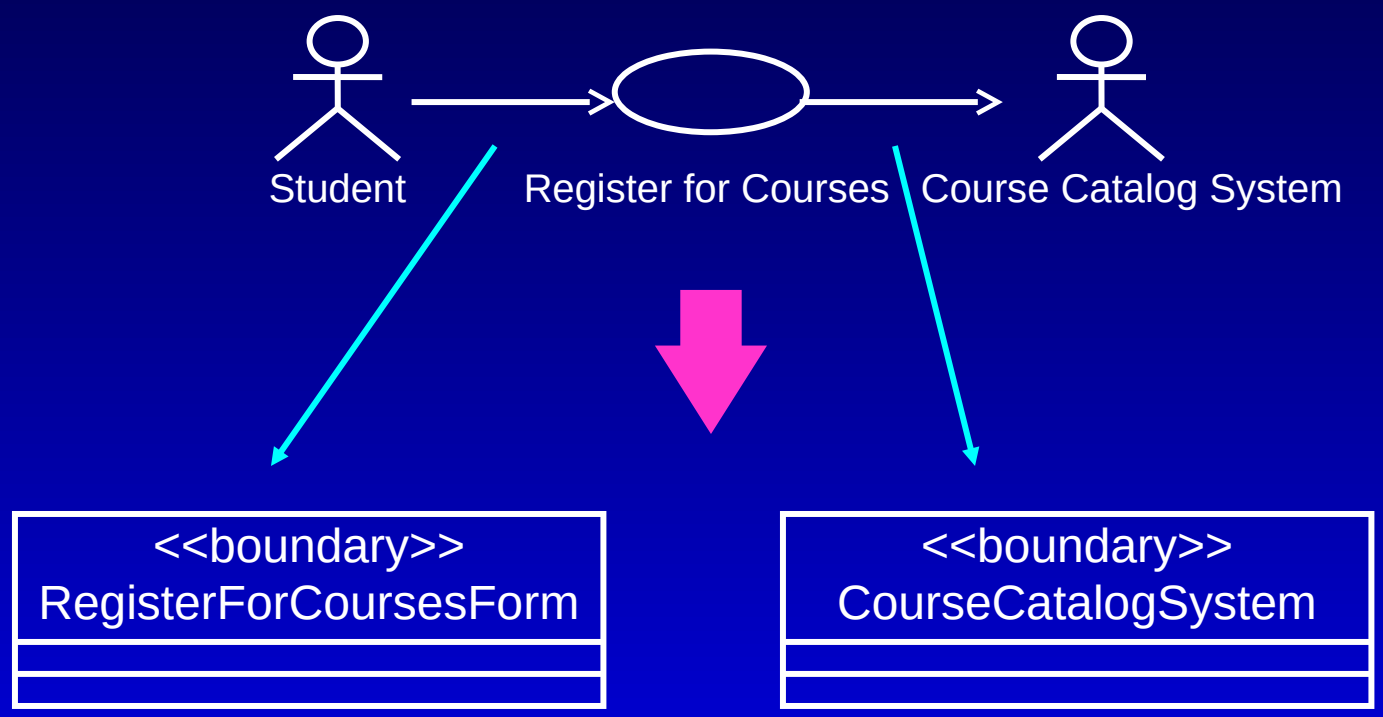
Example: Finding Boundary Classes

- One boundary class per actor/use case pair



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Guidelines: Boundary Class

- User Interface Classes
  - Concentrate on what information is presented to the user
  - Do NOT concentrate on the UI details
- System and Device Interface Classes
  - Concentrate on what protocols must be defined
  - Do NOT concentrate on how the protocols will be implemented

Concentrate on the responsibilities, not the details!

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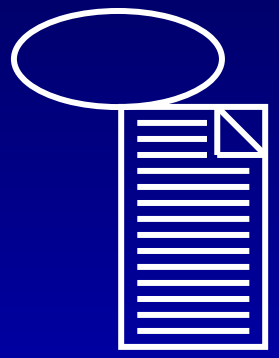
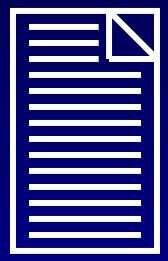
What is an Entity Class?

- Key abstractions of the system

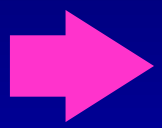


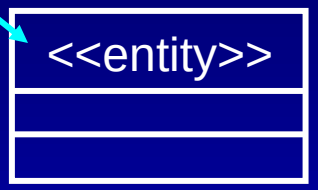
Use Case

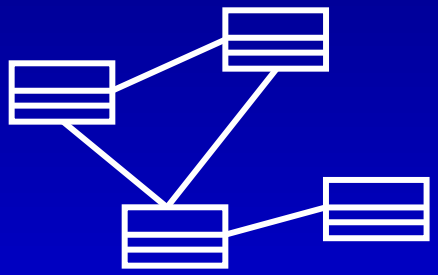


Glossary

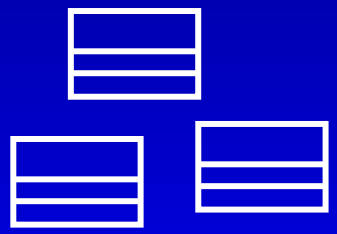
Analysis class stereotype







Business-Domain Model



Architectural Analysis Abstractions

Environment Independent

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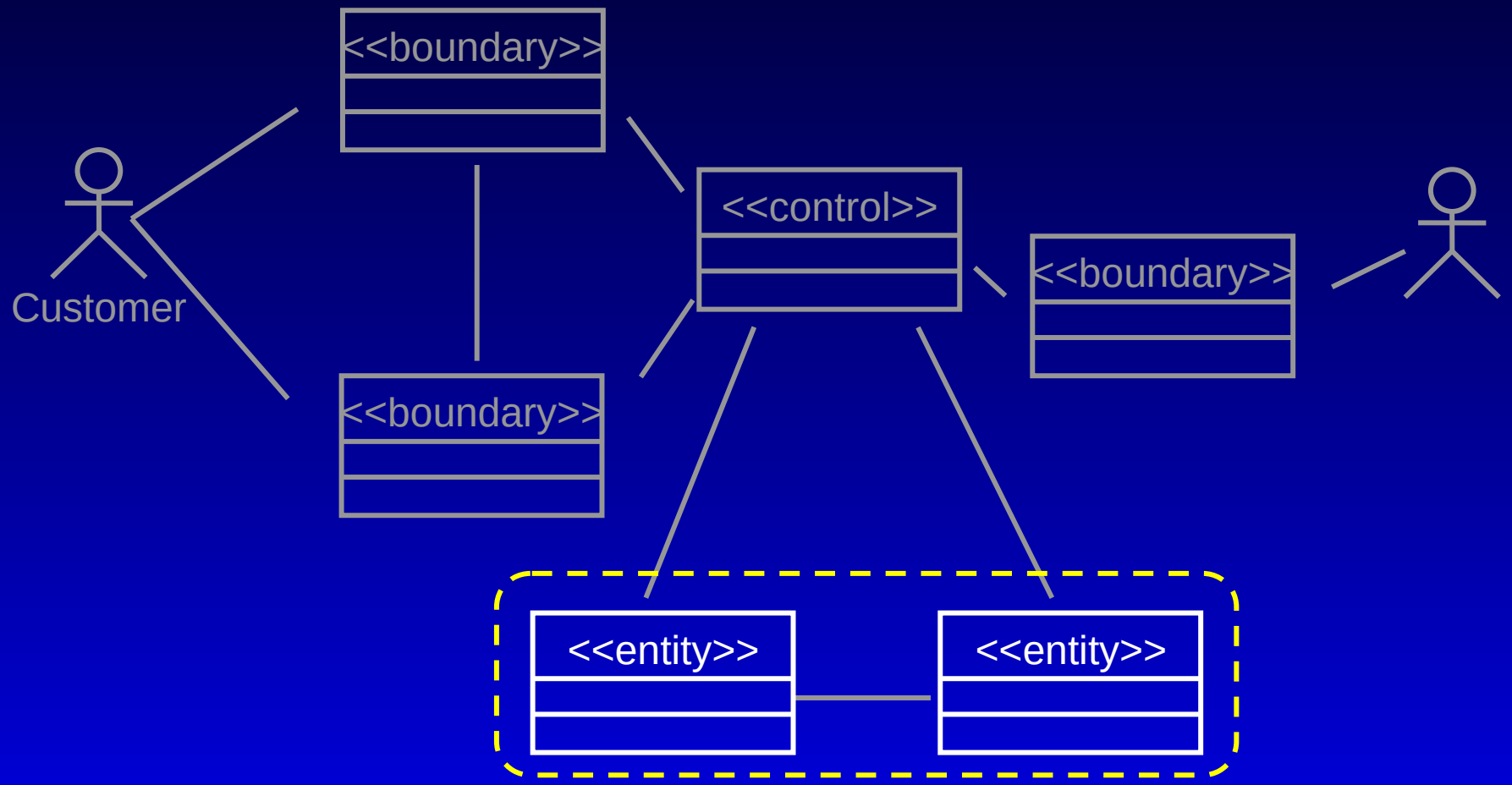
The Role of an Entity Class



Store and manage information in the system

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Example: Finding Entity Classes

- Use use-case flow of events as input
- Key abstractions of the use case
- Traditional, filtering nouns approach
  - Underline noun clauses in the use-case flow of events
  - Remove redundant candidates
  - Remove vague candidates
  - Remove actors (out of scope)
  - Remove implementation constructs
  - Remove attributes (save for later)
  - Remove operations

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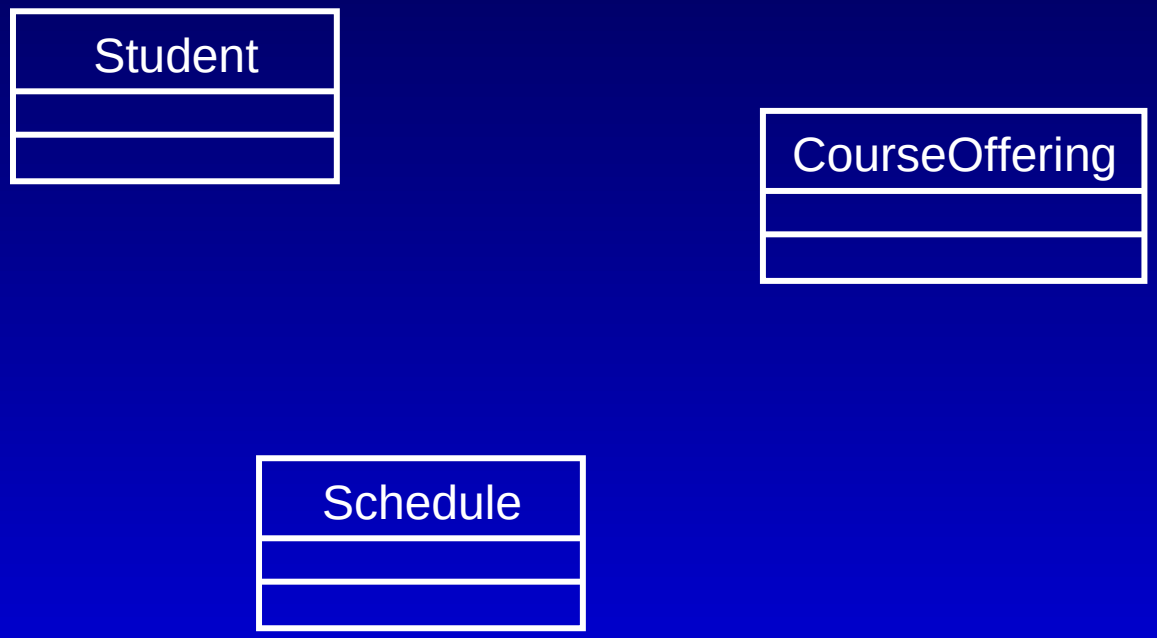
Example: Candidate Entity Classes

- Register for Courses (Create Schedule)



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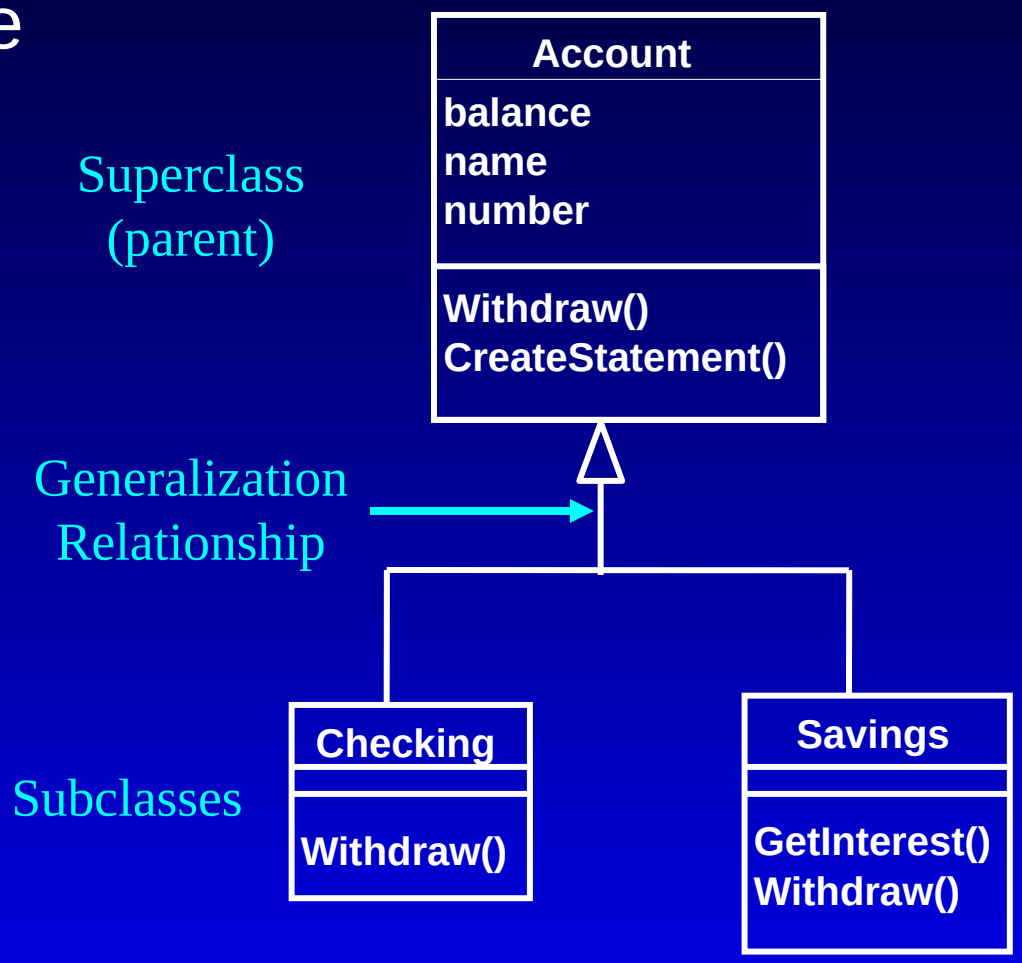
Review: Generalization

- One class shares the structure and/or behavior of one or more classes
- "Is-a-kind of" relationship
- In analysis, use sparingly



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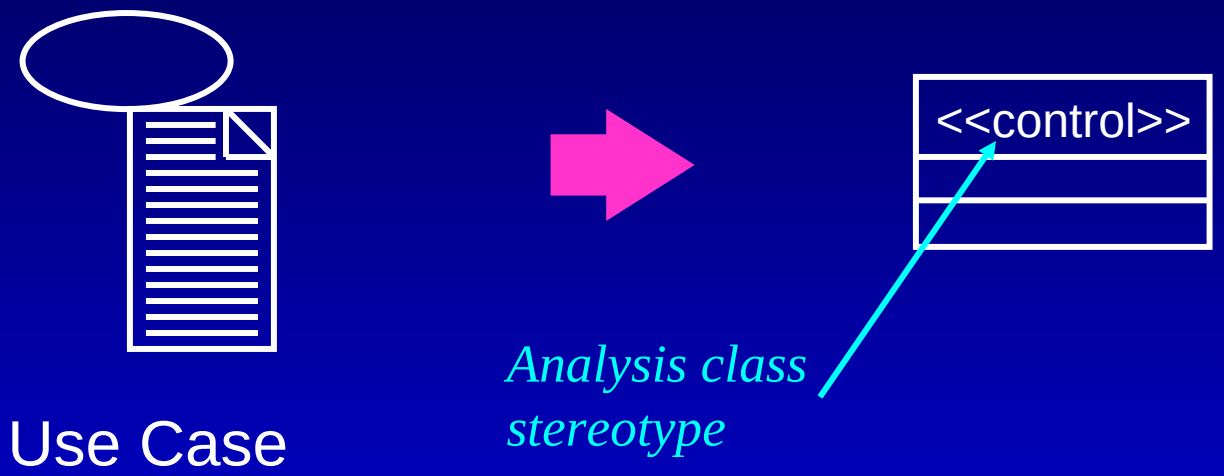
What is a Control Class?

- Use-case behavior coordinator
- *One control class per use case*



Use-case dependent, Environment independent

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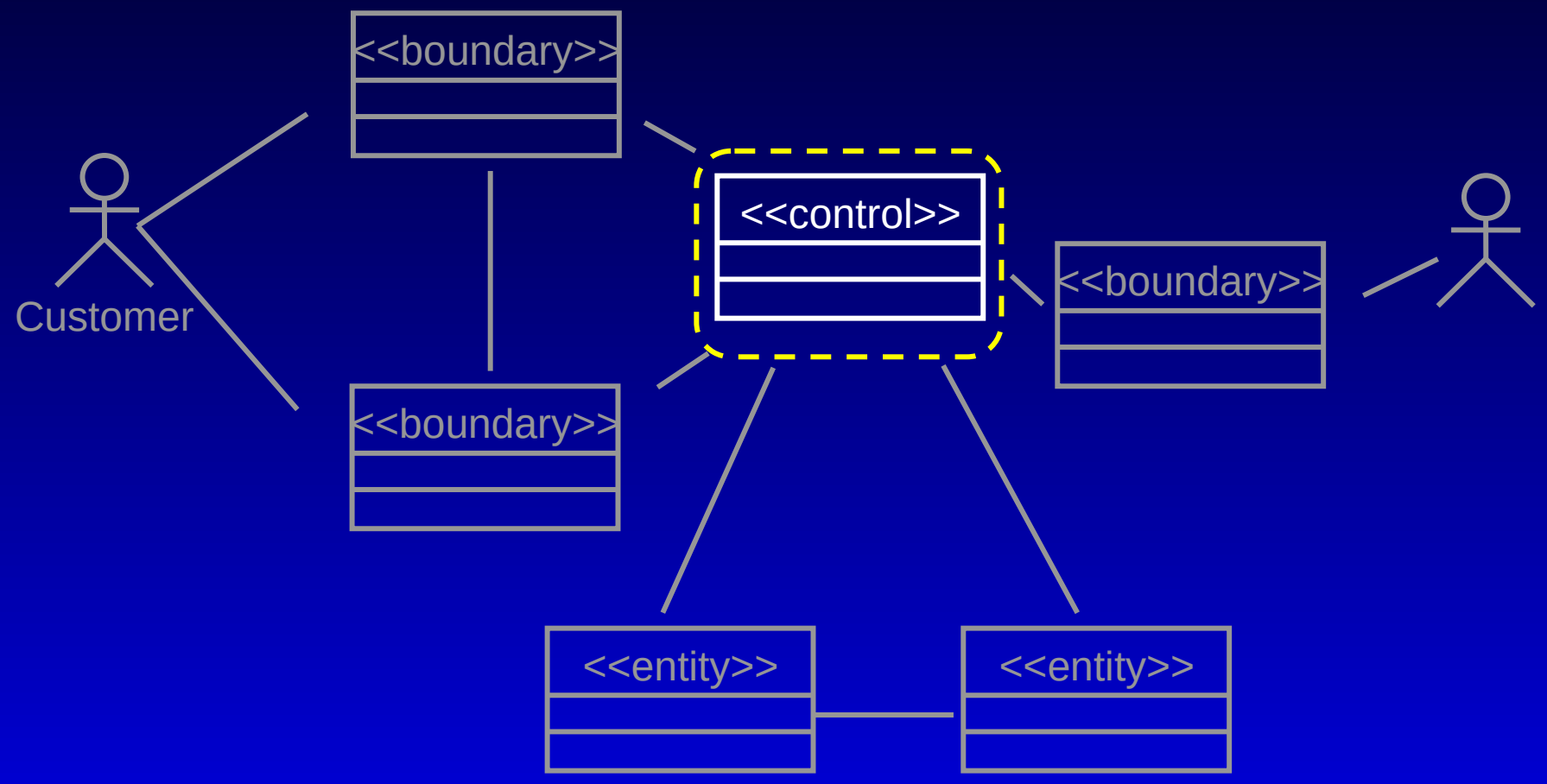
The Role of a Control Class



Coordinate the use-case behavior

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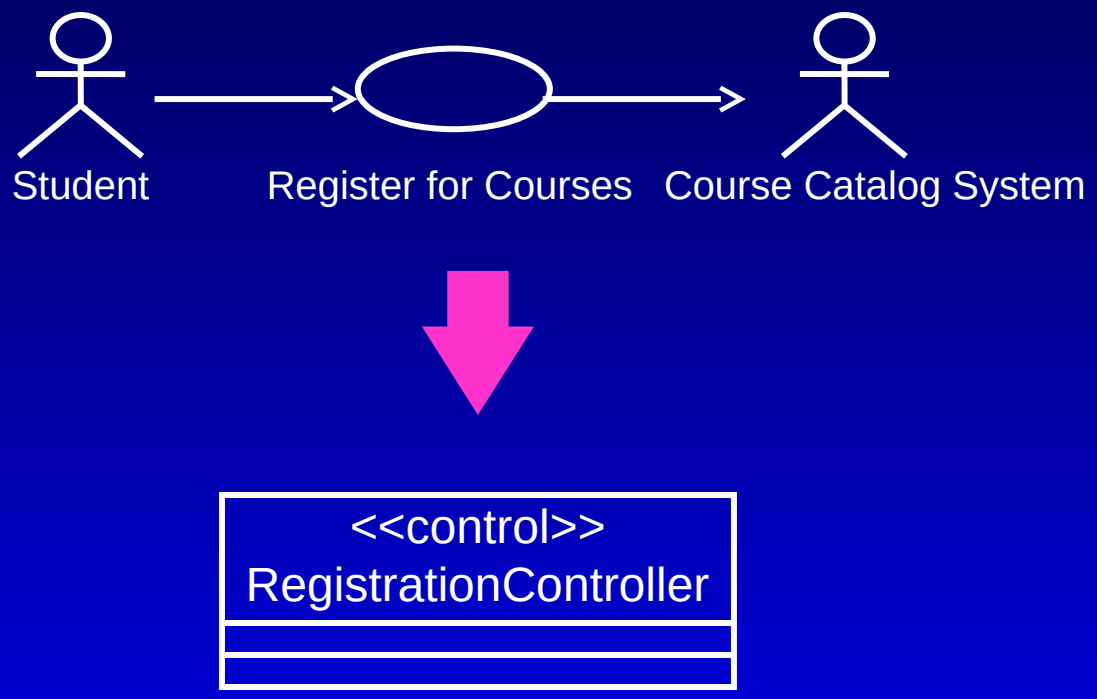
Example: Finding Control Classes

- One control class per use case



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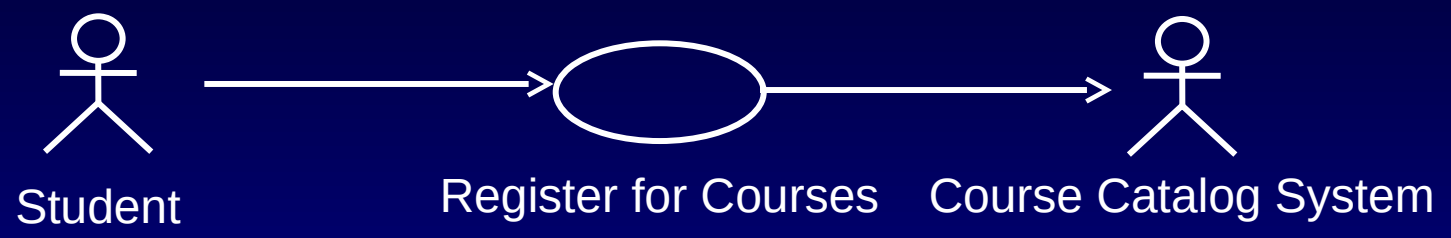
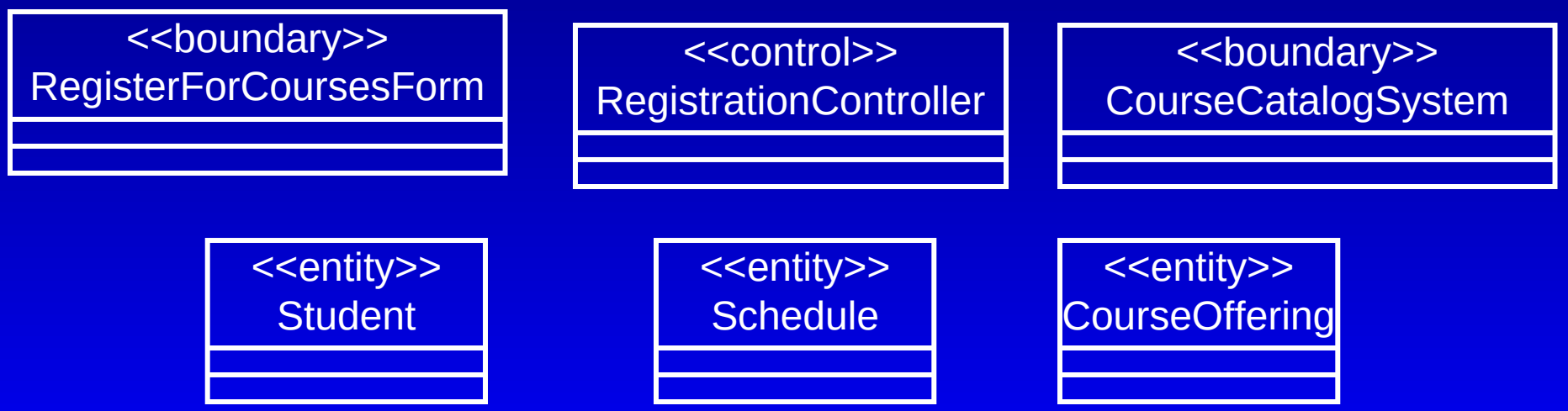
Example: Summary: Analysis Classes



Use-Case Model

Design Model



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Use-Case Analysis Steps

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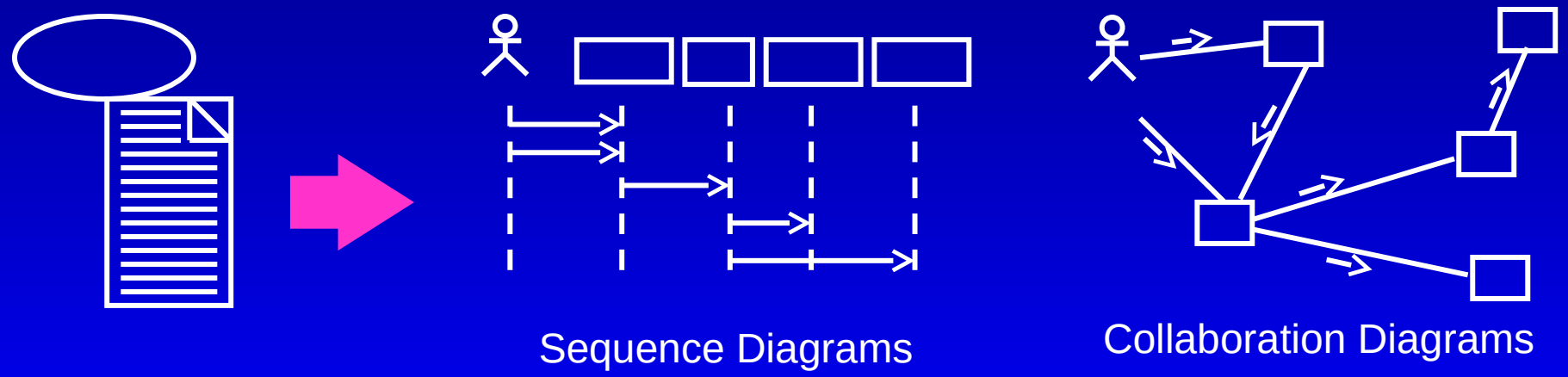
Distribute Use-Case Behavior to Classes

- For each use-case flow of events:
  - Identify analysis classes
  - Allocate use-case responsibilities to analysis classes
  - Model analysis class interactions in interaction diagrams



Use Case

Use-Case Realization

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Guidelines: Allocating Responsibilities to Classes

- Use analysis class stereotypes as a guide
  - Boundary Classes
    - Behavior that involves communication with an actor
  - Entity Classes
    - Behavior that involves the data encapsulated within the abstraction
  - Control Classes
    - Behavior specific to a use case or part of a very important flow of events

(continued)

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Guidelines: Allocating Responsibilities to Classes (cont.)

- Who has the data needed to perform the responsibility?
  - One class has the data, put the responsibility with the data
  - Multiple classes have the data:
    - Put the responsibility with one class and add a relationship to the other
    - Create a new class, put the responsibility in the new class, and add relationships to classes needed to perform the responsibility
    - Put the responsibility in the control class, and add relationships to classes needed to perform the responsibility

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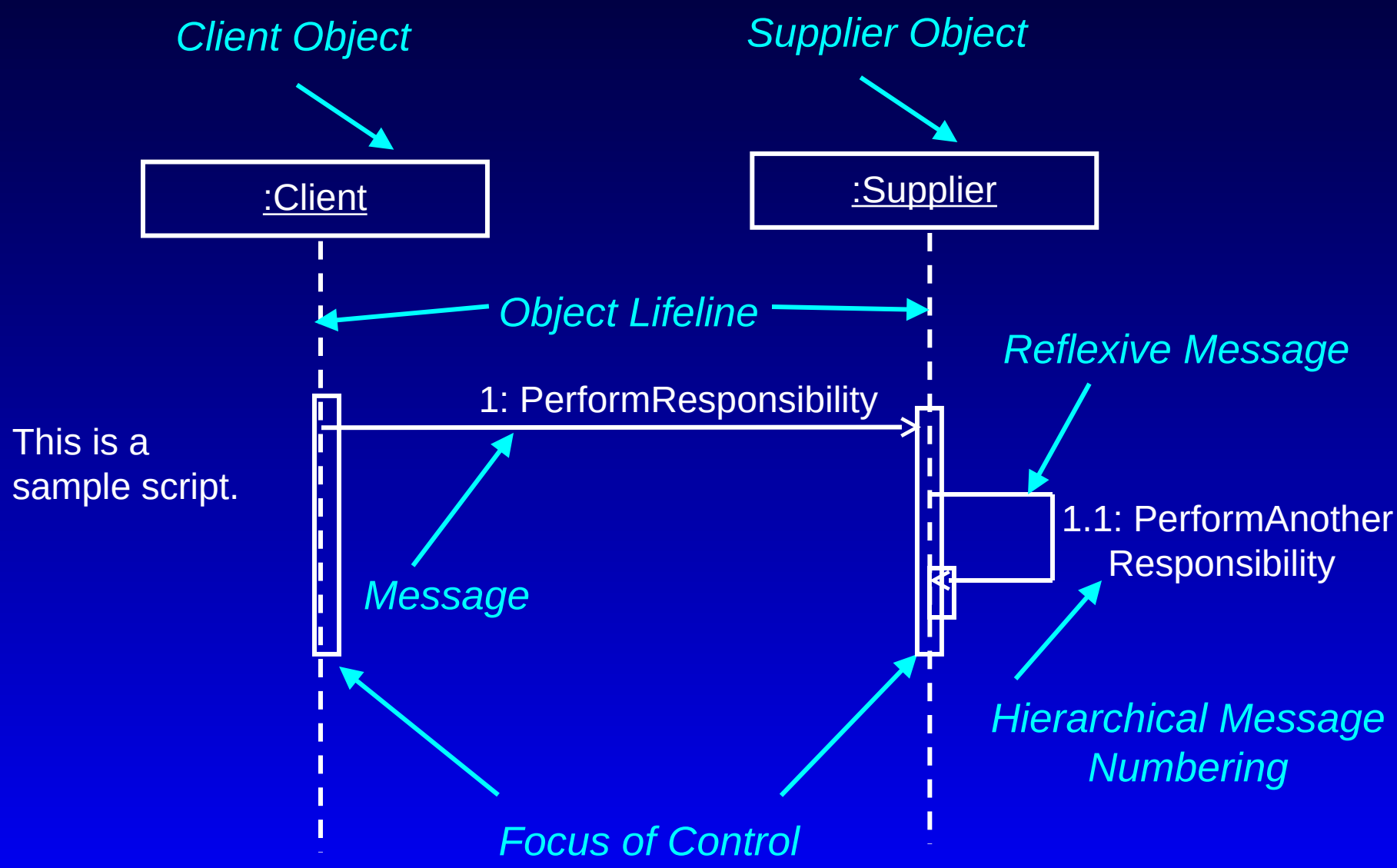
The Anatomy of Sequence Diagrams



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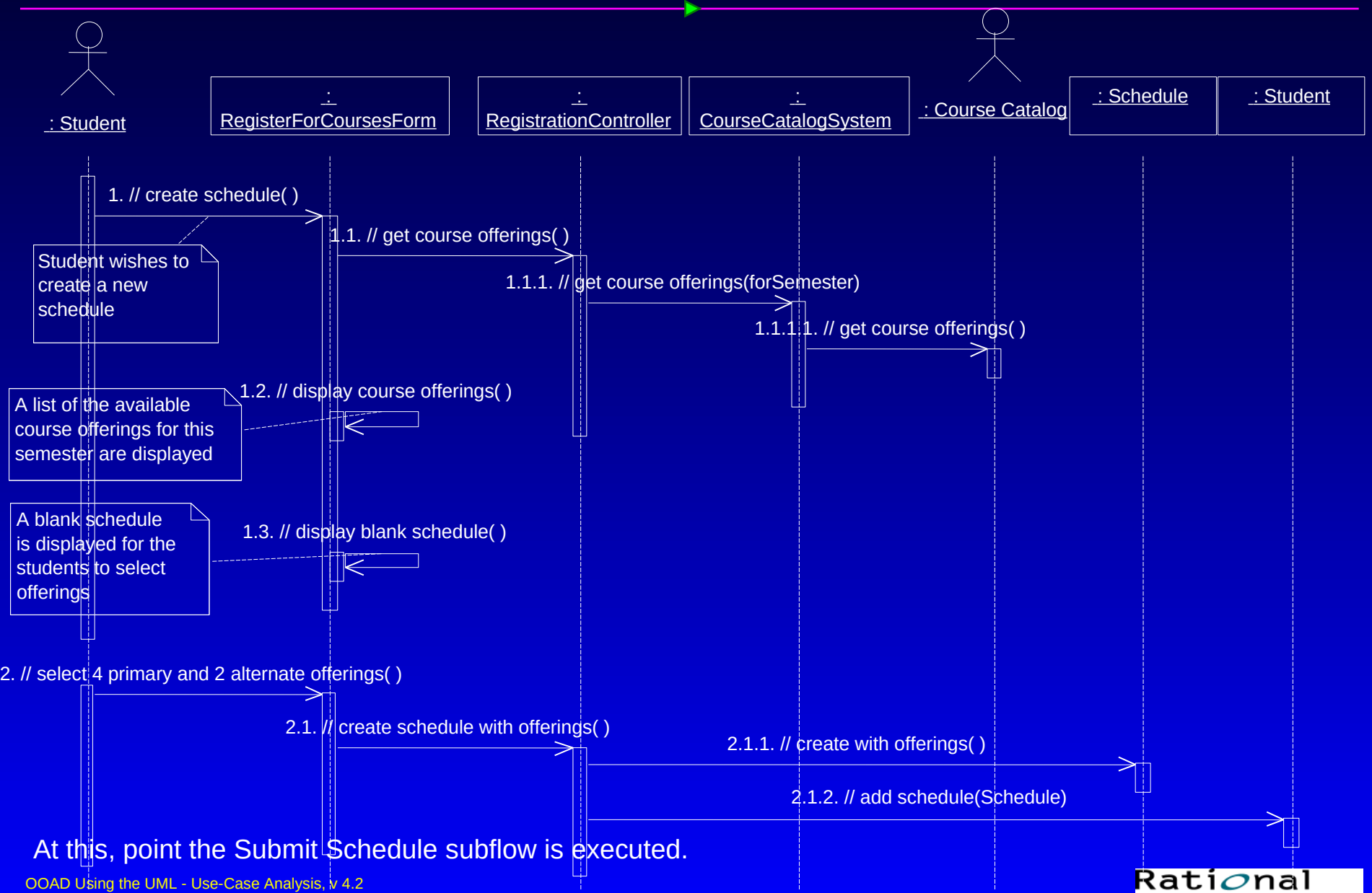
Example: Sequence Diagram



At this, point the Submit Schedule subflow is executed.

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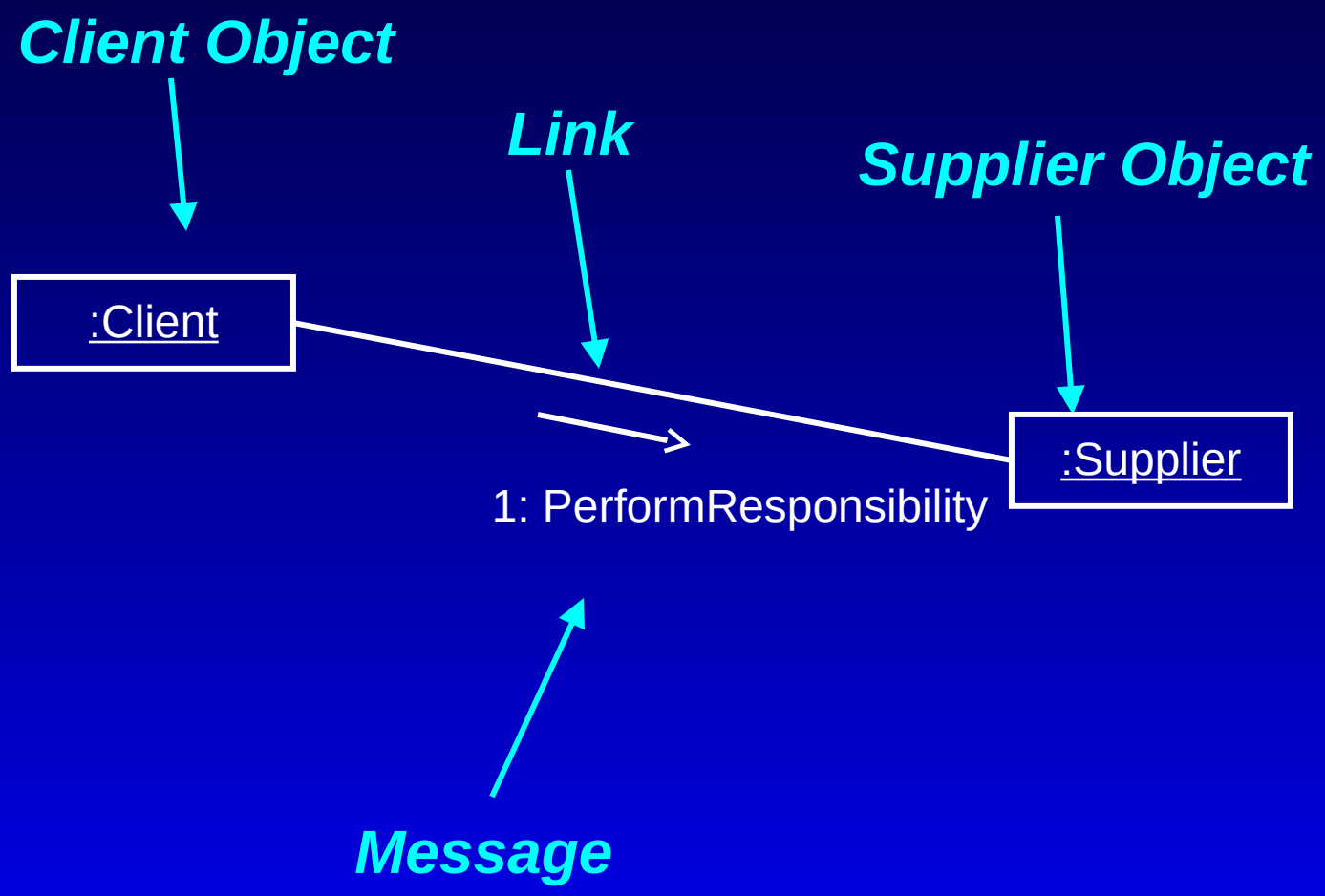
The Anatomy of Collaboration Diagrams



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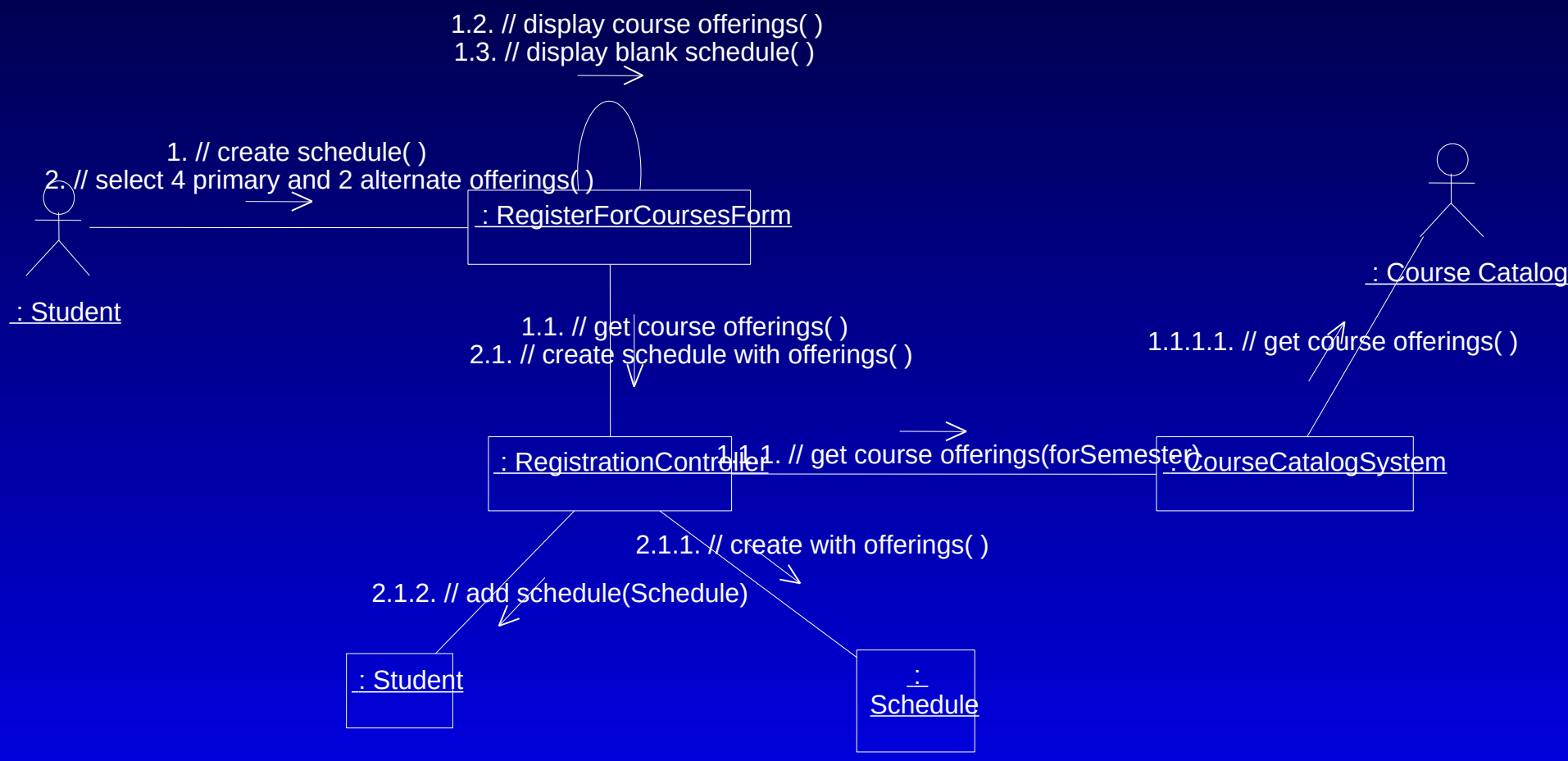
Example: Collaboration Diagram



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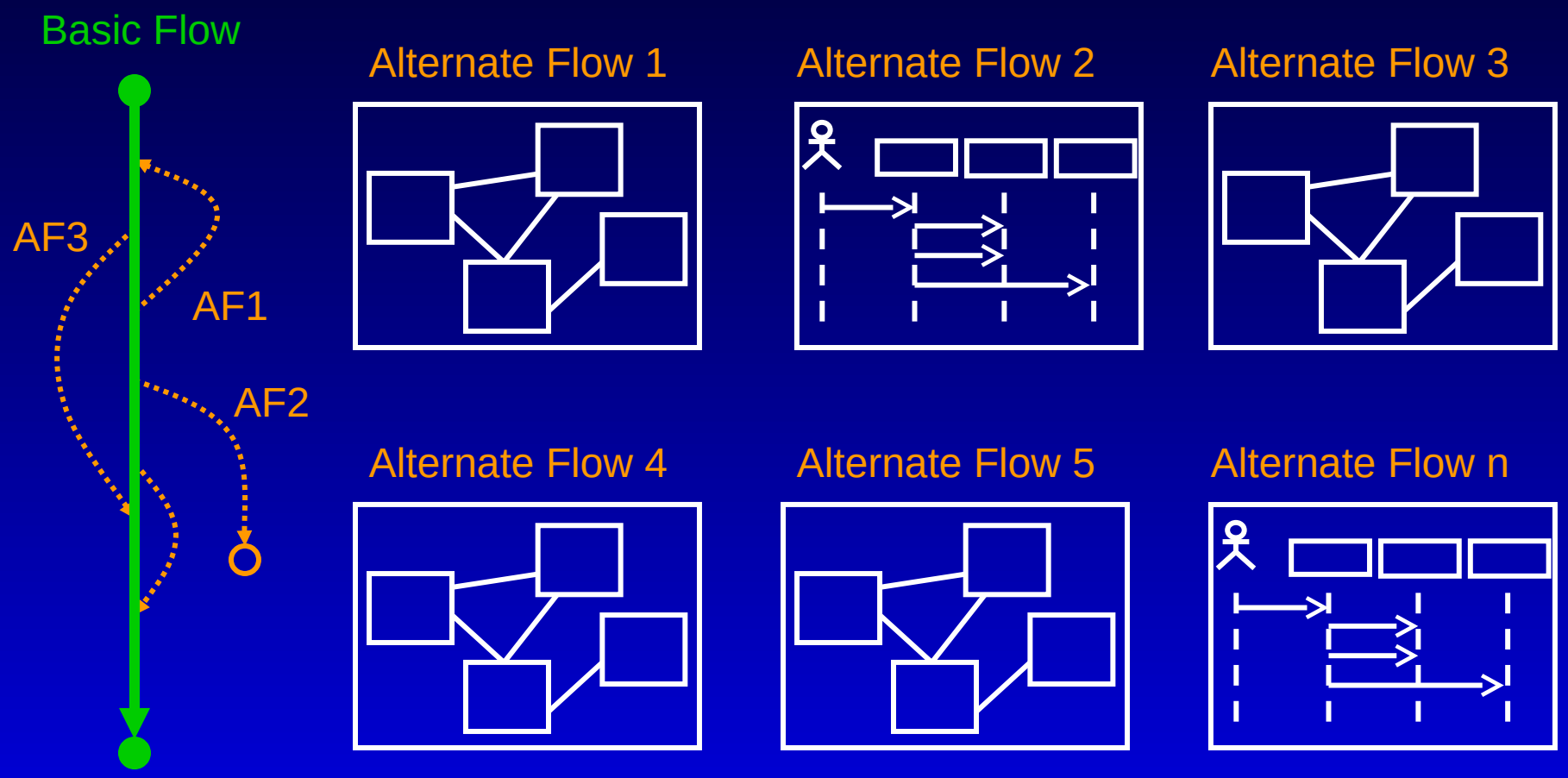
One Interaction Diagram Not Good Enough



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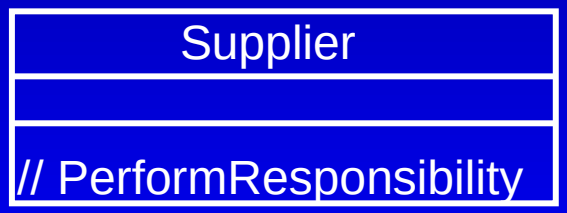
Describe Responsibilities

- What are responsibilities?
- How do I find them?

Interaction Diagram

Class Diagram



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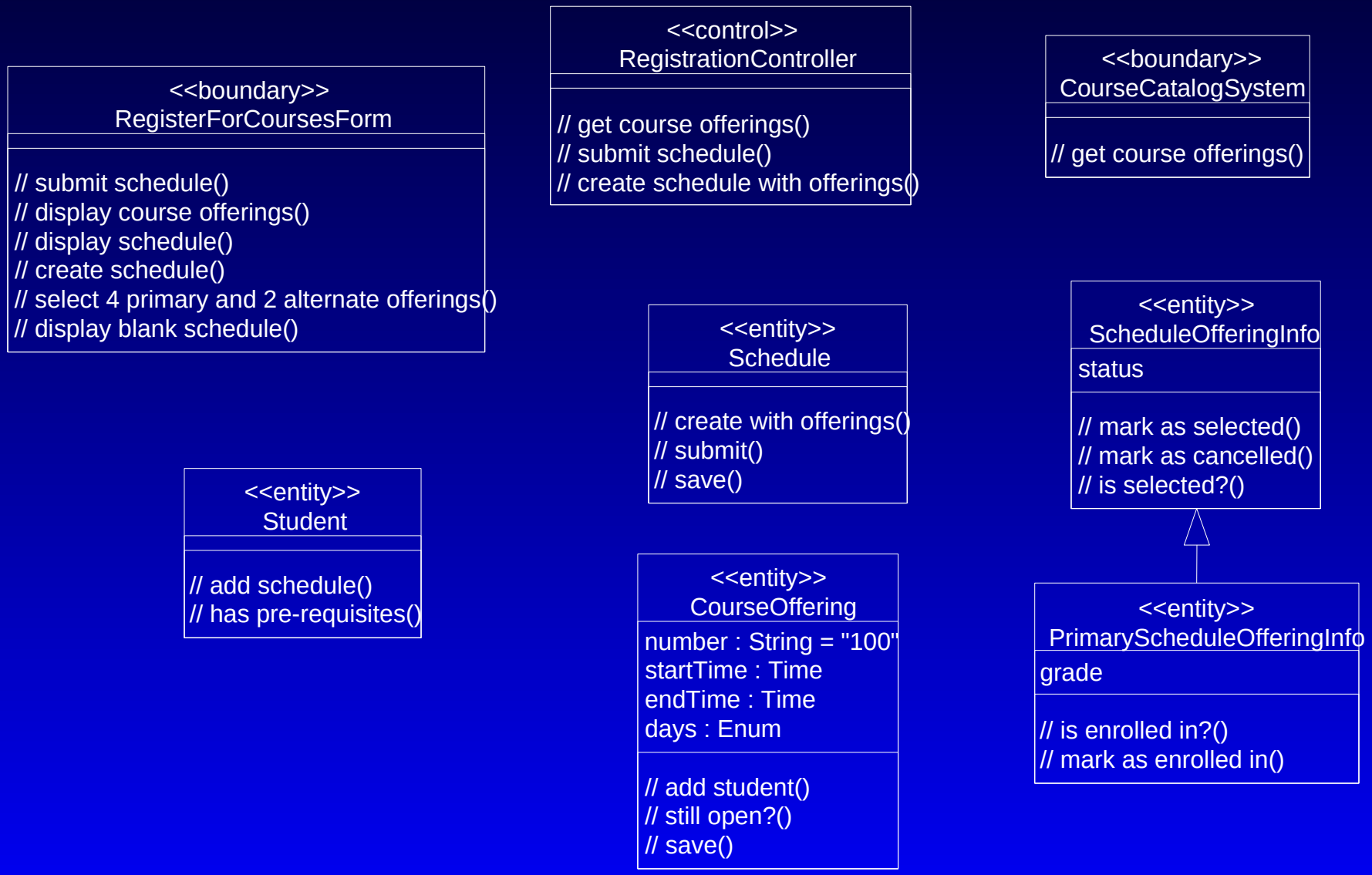
Example: View of Participating Classes (VOPC) Class Diagram



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Describe Attributes and Associations

- Define Attributes
- Establish Aggregations and Associations

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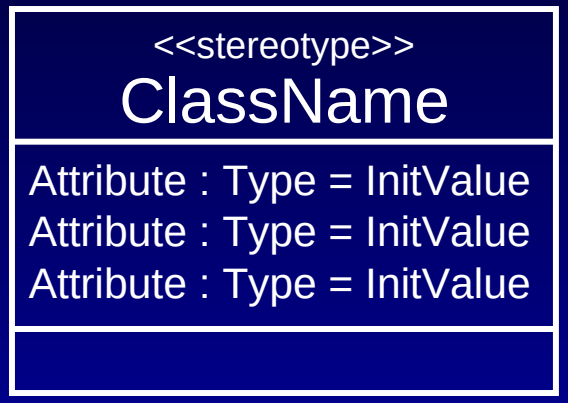
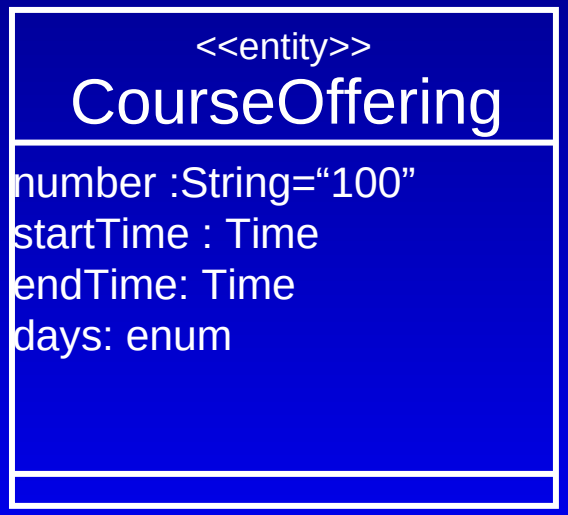
Review: What is an Attribute?



In analysis, do not spend time on attribute signatures



attribute

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Finding Attributes

- Properties/characteristics of identified classes
- Information retained by identified classes
- “Nouns” that did not become classes
  - Information whose value is the important thing
  - Information that is uniquely "owned” by an object
  - Information that has no behavior

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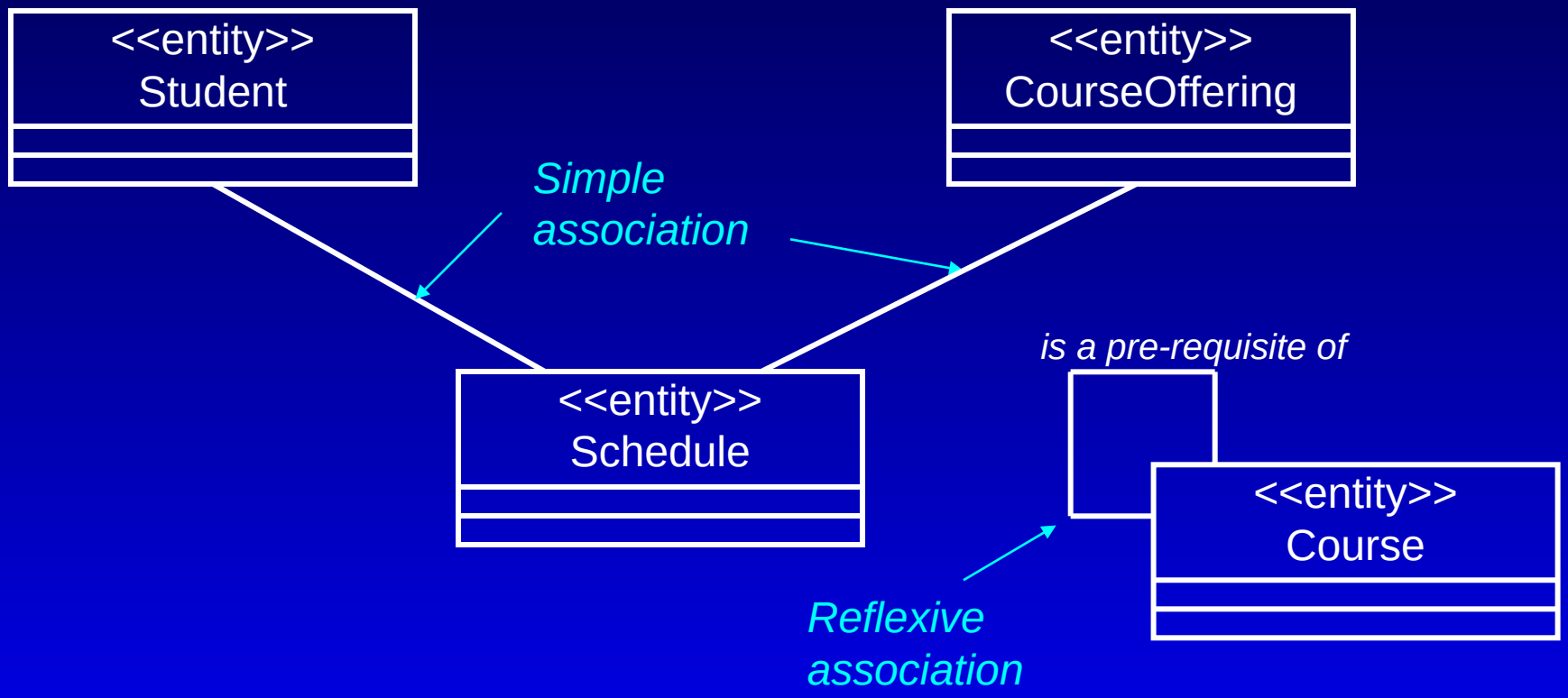
Review: What is an Association?

- Models a semantic connection among instances



Association is a structural relationship

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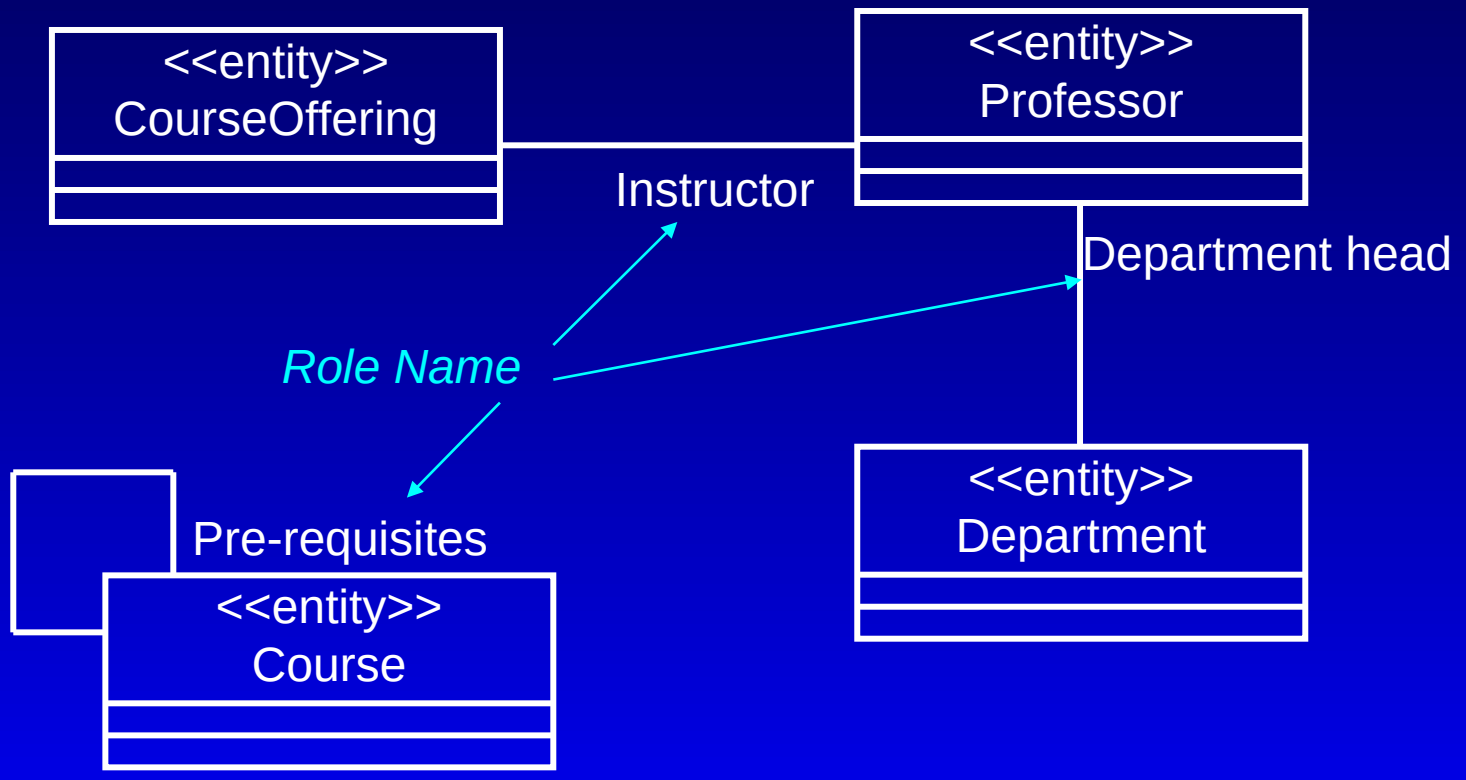
Review: What are Roles?

- The “face” that a class plays in the association



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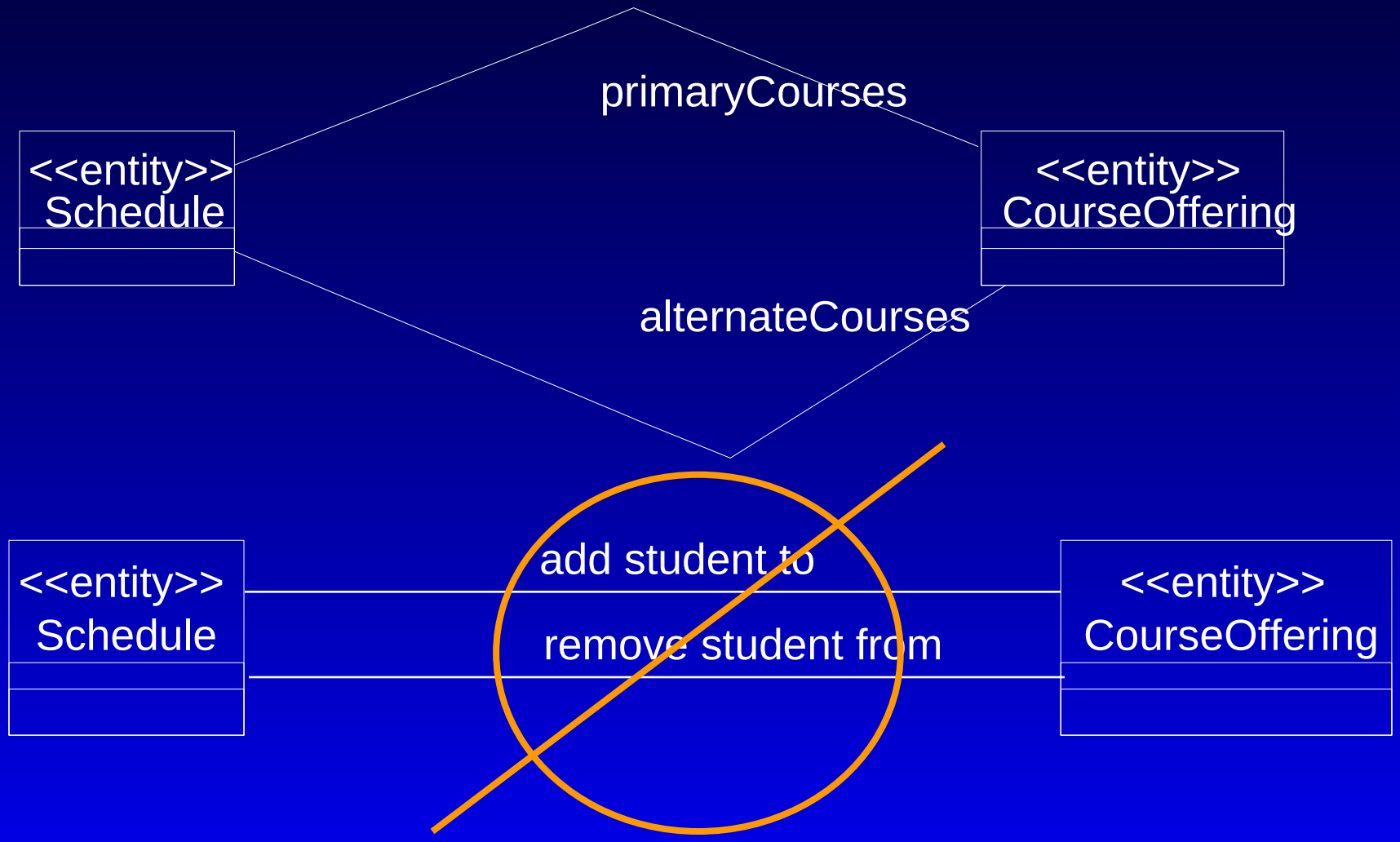
Example: Multiple Associations



Multiple associations must reflect multiple roles

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Review: Multiplicity

- Unspecified
- Exactly one
- Zero or more (many, unlimited)
- One or more
- Zero or one (optional scalar role)
- Specified range
- Multiple, disjoint ranges

1

0..*

*

1..*

0..1

2..4

2, 4..6

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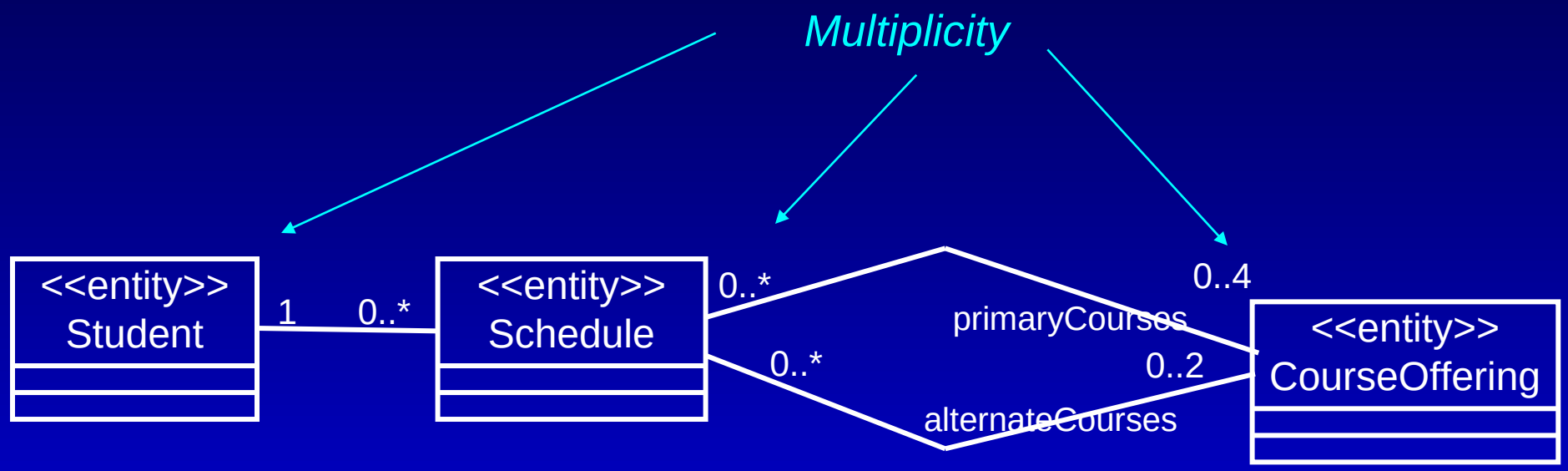
Example: Multiplicity



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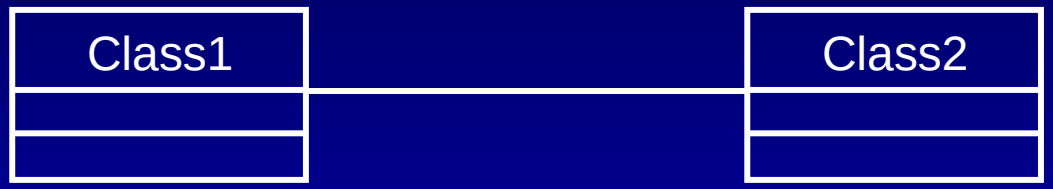
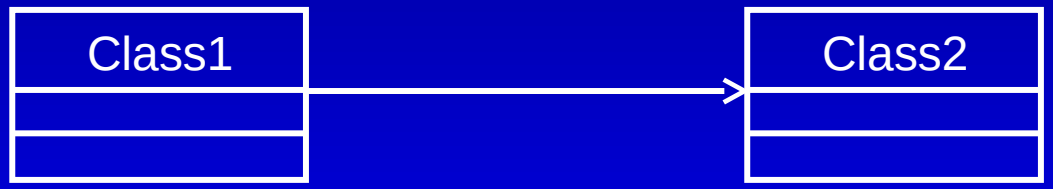
Review: Navigability

- Possible to navigate from an associating class to the target class

Bi-directional



Uni-directional



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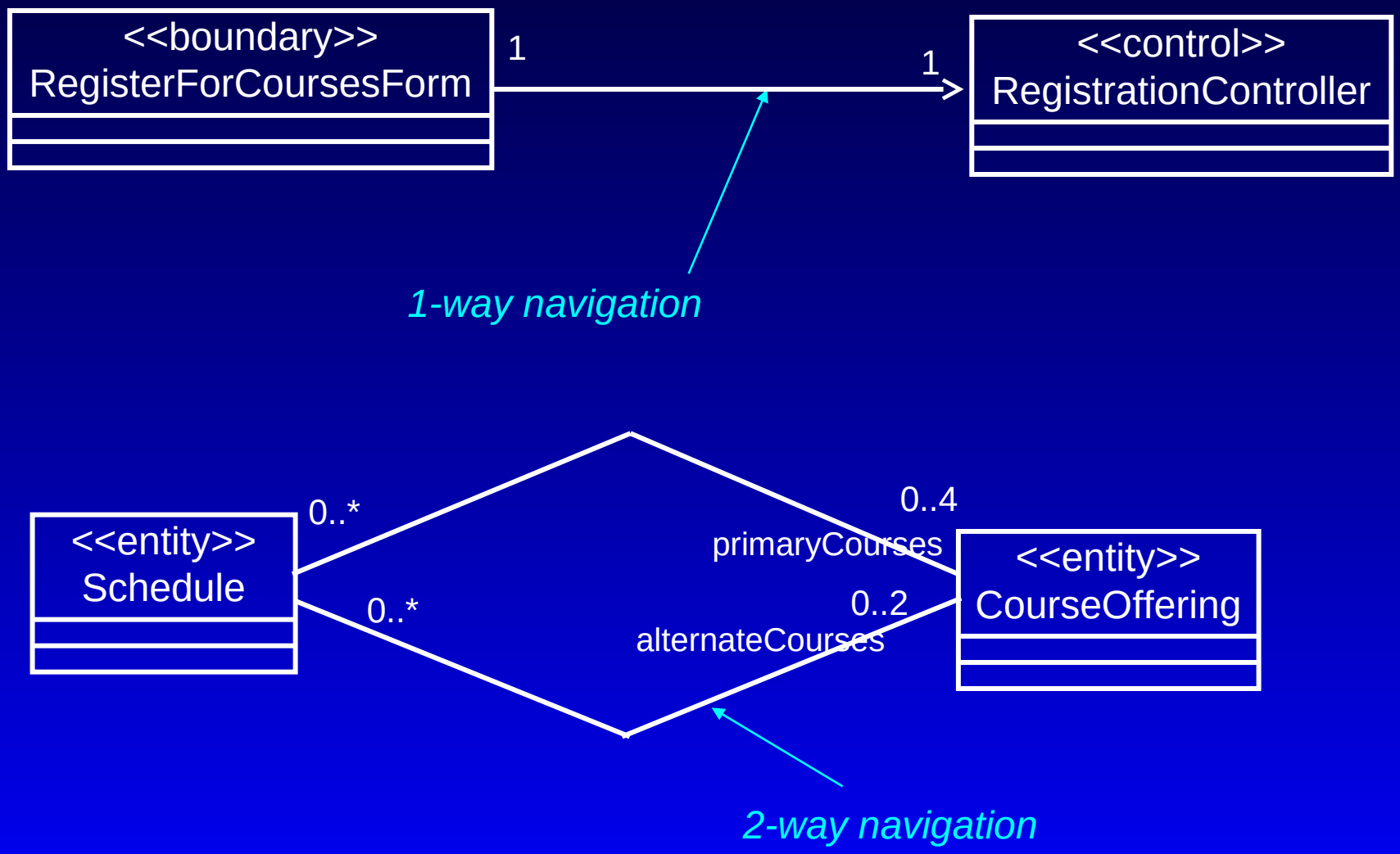
Example: Navigability



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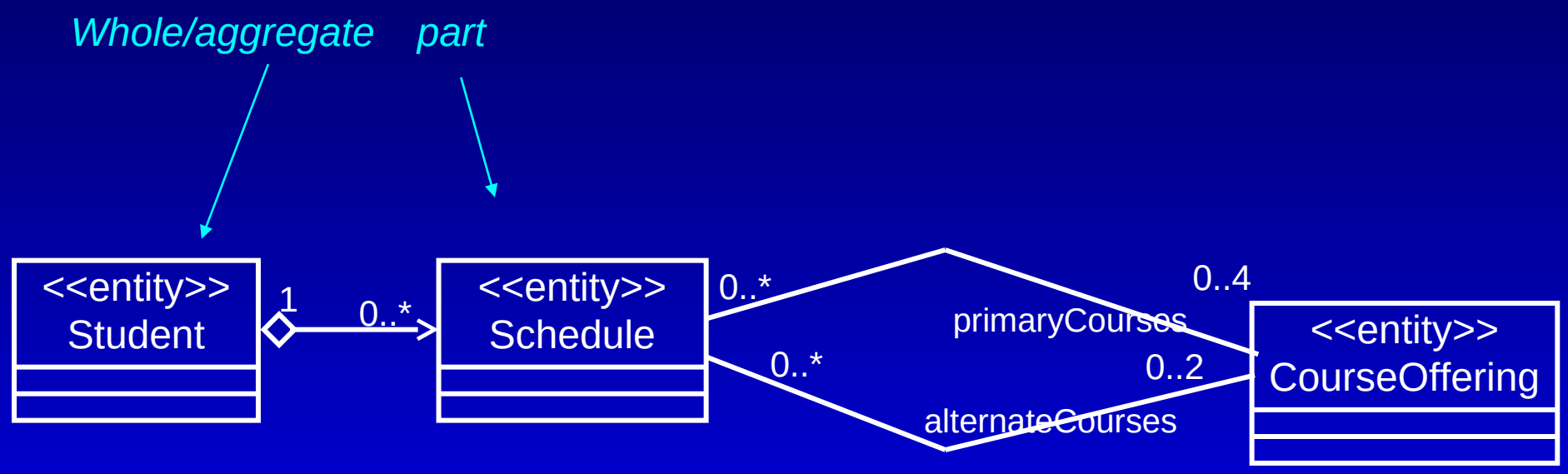
Review: What is Aggregation?

- A special form of association that models a whole-part relationship between an aggregate (the whole) and its parts



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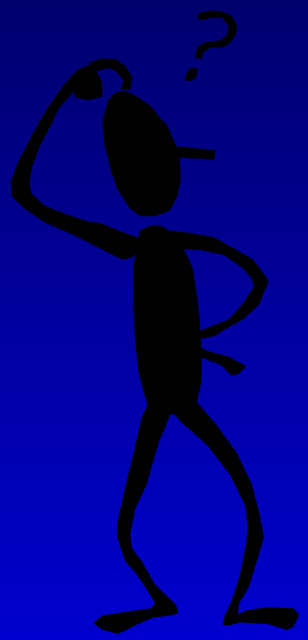
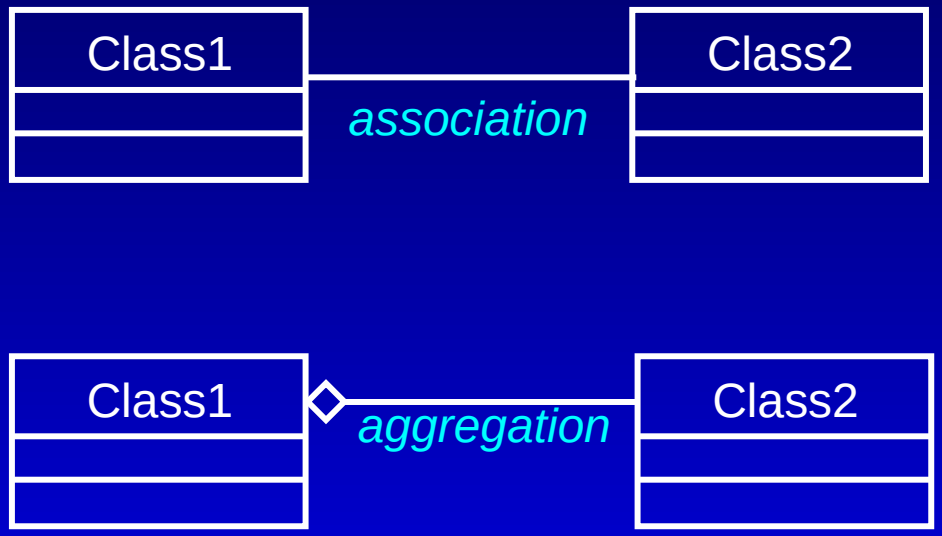
Association or Aggregation?

- Consideration
  - Context, independent identity of Class2





When in doubt use association

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Finding Relationships

Collaboration Diagram

1: PerformResponsibility

:Client

:Supplier

Link

Client

Supplier

Class Diagram

Client

0..*

0..*

Prime suppliers

Supplier

PerformResponsibility()

Association

Relationship for every link!

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Use-Case Analysis Steps

- Supplement the Use-Case Descriptions
- For each use-case realization
  - Find Classes from Use-Case Behavior
  - Distribute Use-Case Behavior to Classes
- For each resulting analysis class
  - Describe Responsibilities
  - Describe Attributes and Associations
  - Qualify Analysis Mechanisms
- Unify Analysis Classes
- Checkpoints

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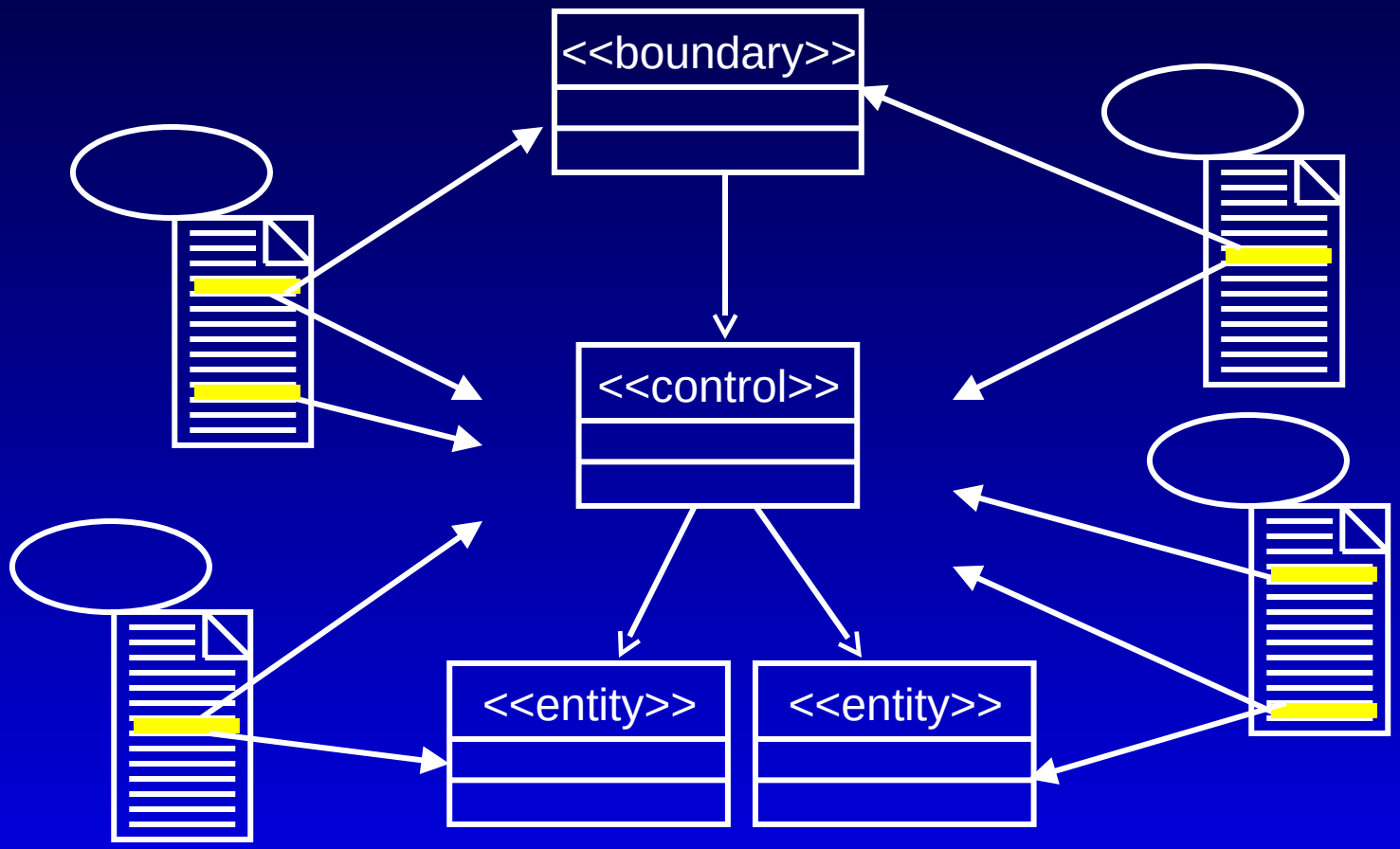
Unify Analysis Classes



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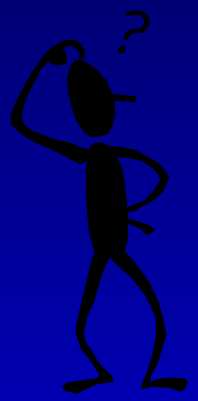
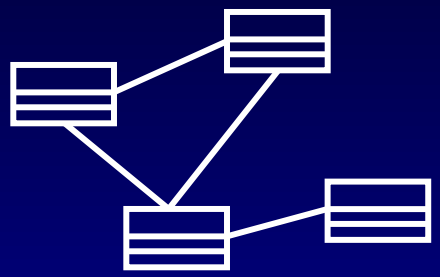
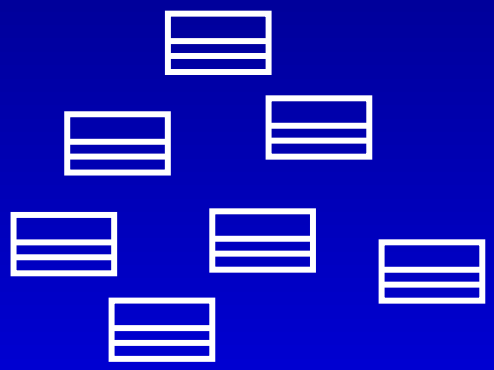
Evaluate Your Results

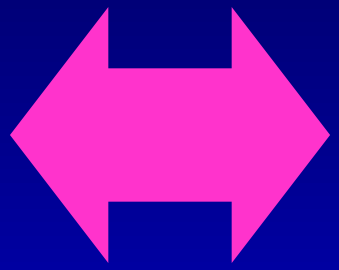




Design Model

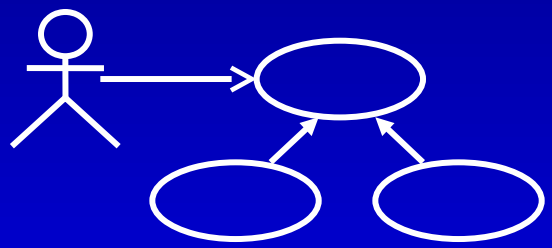


Analysis Classes



Supplementary Specification

Glossary



Use-Case Model

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Use-Case Analysis Steps

- Supplement the Use-Case Descriptions
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- For each resulting analysis class
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  - Describe Attributes and Associations
  - Qualify Analysis Mechanisms
- Unify Analysis Classes
- Checkpoints

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Checkpoints: Analysis Classes

- Are the classes reasonable?
- Does the name of each class clearly reflect the role it plays?
- Does the class represent a single well-defined abstraction?
- Are all attributes and responsibilities functionally coupled?
- Does the class offer the required behavior?
- Are all specific requirements on the class addressed?

(continued)

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Checkpoints: Use-Case Realizations

- Have all the main and/or sub-flows been handled, including exceptional cases?
- Have all the required objects been found?
- Has all behavior been unambiguously distributed to the participating objects?
- Has behavior been distributed to the right objects?
- Where there are several interaction diagrams, are their relationships clear and consistent?

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Review: Use-Case Analysis

- What is the purpose of Use-Case Analysis?
- What is an analysis class? Name and describe the three analysis stereotypes.
- What is a use-case realization?
- Describe some considerations when allocating responsibilities to analysis classes.
- How many interaction diagrams should be produced during Use-Case Analysis?

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